

DEBRA HOGUE, PhD

Computer Scientist · Human-AI Collaboration Researcher

Oklahoma City Metro Area · contact@debrahogue.phd · linkedin.com/in/dr-debra-hogue-14653026 · debrahogue.phd

EDUCATION

Ph.D., Computer Science <i>University of Oklahoma · Gallogly College of Engineering · Norman, OK</i> <i>Dissertation: "From Transparency to Agency: The MICA Framework for Addressing the Understanding-Trust Gap in AI-Assisted Camouflage Analysis." Committee: Dr. Ghulam Jilani Quadri (Chair), Dr. Dean F. Hougen, Dr. Sridhar Radhakrishnan, Dr. Justin G. Metcalf. DoD SMART Scholar.</i> Available on ShareOK: shareok.org/handle/11244/342777	2026
M.S., Computer Science <i>Oklahoma Christian University</i>	2017
B.S., Computer Science <i>University of Central Oklahoma</i>	2015

RESEARCH PROJECTS

MICA — Mixed-Initiative Camouflage Analysis Framework <i>University of Oklahoma · DIV-Lab · Oct 2024–Mar 2026 · Advisor: Dr. G.J. Quadri</i> Advanced MURDOC into a full mixed-initiative framework enabling bidirectional human-AI collaboration through ROI selection, adjustable sensitivity, and user feedback mechanisms, in response to findings that visual explanations alone improved understanding but did not build sufficient trust.	
AI-Generated Camouflage Synthesis using Deep Learning <i>University of Oklahoma · Gallogly College of Engineering · Dec 2024–Mar 2026</i> Developed a specialized machine learning model using Low-Rank Adaptation (LoRA) fine-tuning to generate photorealistic camouflaged objects across biological and non-biological categories, addressing dataset scarcity in camouflage research.	
MURDOC — Transforming Pixels into Perception <i>IEEE DASC 2024 · Feb 2023–Oct 2024</i> Interactive offline visualization tool integrating Informative AI (IAI) and eXplainable AI (XAI) techniques for camouflage detection, using a three-level decision hierarchy (ResNet50, EfficientDet-D7), Grad-CAM attention visualization, and the TOAST trust-measurement scale.	
Video Motion Extraction for Temporal Analysis <i>University of Oklahoma · Gallogly College of Engineering · Jan 2024–Sep 2024</i> Explored temporal motion patterns in camouflaged wildlife using Porter-Duff blend modes and mixed-abstraction visual encoding; informed the IEEE VIS 2024 poster and related publication.	
The Crisis Adoption Problem <i>Thought Leadership · Conference Pipeline · 2026</i> Original theoretical concept extending dissertation findings on time pressure and AI over-reliance into a structural critique of institutional AI adoption.	

PUBLICATIONS

Building Operator Trust Through Mixed-Initiative Camouflage Analysis for Surveillance <i>Hogue, D., Elliott, D.S., & Quadri, G.J. · IEEE 45th Digital Avionics Systems Conference (DASC), Orlando, FL</i>	2026 · Accepted, Publication Pending
Multi-Architecture Evaluation of Stable Diffusion with HSV–Gestalt Perceptual Loss <i>Hogue, D., Elliott, D.S., & Quadri, G.J. · IEEE Computer Graphics and Applications</i>	Under Review
From Transparency to Agency: The MICA Framework for Addressing the Understanding-Trust Gap in AI-Assisted Camouflage Analysis <i>Doctoral Dissertation · University of Oklahoma, Gallogly College of Engineering</i> Available on ShareOK: shareok.org/handle/11244/342777	2026

Human-AI Trust Calibration in Mixed-Initiative Systems: A Framework for High-Stakes Environments

AIAA/IEEE DASC 2025 · Montreal, Canada

2025

MURDOC: Transforming Pixels into Perception for Camouflage Detection

IEEE 43rd Digital Avionics Systems Conference (DASC)

2024 · Best in Session

Using Informative AI to Understand Camouflaged Object Detection and Segmentation

IEEE/AIAA 42nd Digital Avionics Systems Conference (DASC)

2023 · Best in Session

CONFERENCE PRESENTATIONS & TALKS

Building Operator Trust Through Mixed-Initiative Camouflage Analysis for Surveillance

IEEE 45th DASC · Orlando, FL

2026 · Accepted, Presentation Pending

Human-AI Collaboration: Trust, Uncertainty, and the Design of Intelligent Systems

InnoTech Oklahoma / OKWIT · Panel Speaker

2025

MICA: Trust-Driven Design Refinements for Camouflaged Object Detection Applications

IEEE 44th Digital Avionics Systems Conference (DASC) · Montreal, Canada

Dec 2025 · Best in Session

MURDOC: Transforming Pixels into Perception for Camouflage Detection

IEEE 43rd DASC

Nov 2024 · Best in Session

Using Informative AI to Understand Camouflaged Object Detection and Segmentation

IEEE/AIAA 42nd DASC

Nov 2023 · Best in Session

Visual Analysis of Motion for Camouflaged Object Detection

IEEE VIS 2024 · Poster

Oct 2024 · Best Poster Honorable Mention

AWARDS & HONORS

- ◆ 44th AIAA/DASC Best in Session — IEEE DASC · December 2025 · MICA Framework
- ◆ 43rd AIAA/DASC Best in Session — IEEE DASC · November 2024 · MURDOC
- ◆ 42nd AIAA/DASC Best in Session — IEEE DASC · November 2023 · Informative AI Paper
- ◆ IEEE VIS Best Poster Honorable Mention — IEEE VIS 2024 · Visual Analysis of Motion
- ◆ SMART Retention Scholar — DoD Science, Mathematics & Research for Transformation